

Programme	B. Sc. Mathematics Honours				
Course Code	MAT6FS113(1)				
Course Title	DATA SCIENCE WITH PYTHON				
Type of Course	SEC (for pathways 1 – 5)				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	0	45
Pre-requisites	A basic course in Python programming with the understanding of using looping, conditionals, creating variables, writing functions, and importing modules.				
Course Summary	This course is an advanced course for those who have learned the basics of Python. It will enable the students to learn more features of Python with a specific focus on how to use them to analyse data and arrive at conclusions in practical situations with the help of a reasonable knowledge of statistics.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Learn to rearrange and manipulate various data structures in Python to make it more meaningful	U	F	Internal Exam/ Assignments / End Semester Examination
CO2	Understand fundamentals of Statistics from a real-life point of view	U	F	Internal Exam/ Assignments / Quiz / End Semester Examination
CO3	Learn how to visualise data for clearer understanding of practical situations	Ap	C	Internal Exam / Quiz / End Semester Examination
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)# - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Note : Python IDLE (with necessary modules like pandas, scipy), Anaconda/Spyder package, Jupyter notebook interface or Google colab (free to use) interface, Pydroid 3 for android (along with Pydroid repository plugin) can be used for training purposes. Python version 3.10 or above should be used to avoid errors with some of the functionalities we discuss in the course.

Textbook		1 Mastering Python for Data Science, Samir Madhavan, PACKT Publishing, 2015 2 Data Science from Scratch, Second Edition ,Joel Grus, O'Reilly, 2019		
Module	Unit	Content	Hrs (36+ 9)	Ext. Marks (50)
I	Python Tools for Handling and Manipulating Data (Text 2, Chapter 2)		8	Min 10
	1	Exceptions, Lists.		
	2	Tuples, Dictionaries.		
	3	Counters, Sets, List Comprehensions,		
	4	Truthiness, Automated Testing and assert Iterables and Generators		
	5	Randomness, Regular Expressions, zip and Argument Unpacking		
II	More Tools for Data Handling – Numpy and Pandas (Text 1, Chapter 1)		8	Min 10
	6	NumPy: Mathematical operations, Array subtraction, squaring an array, A trigonometric function performed on the array, Conditional operations.		
	7	NumPy : Matrix multiplication, Indexing and slicing, Shape manipulation.		

	8	Pandas : Inserting and exporting data, CSV, Data cleansing, Checking the missing data.		
	9	Pandas : Filling the missing data, String operations, Merging data		
	10	Data operations: Aggregation operations, Joins, The inner join		
	11	Data operations: The left outer join, The full outer join, The groupby function		
III	Inferential Statistics (Text 1, Chapter 2)		12	Min 10
	12	Various forms of distribution, A normal distribution, A normal distribution from a binomial distribution.		
	13	A Poisson distribution, A Bernoulli distribution.		
	14	A z-score, A p-value, One-tailed and two-tailed tests.		
	15	Type 1 and Type 2 errors, confidence interval.		
	16	Correlation, Z-test vs T-test, The F distribution.		
	17	The chi-square distribution, Chi-square for the goodness of fit, The chi-square test of independence, ANOVA.		
IV	Applying the Theory to Problems (Text 1, Chapter 3)		8	Min 10
	18	What is data mining? Presenting an analysis.		
	19	Studying the Titanic – with all the required analysis		
V	Open Ended Visualizing Data (Text 1, Chapter 4)		10	
	1	Making Sense of Data through Advanced Visualization - Controlling the line properties of a chart		

	2	Using keyword arguments, Using the setter methods, Using the setp() command.		
	3	Creating multiple plots, Playing with text, Styling your plots.		
	4	Box plots, Heatmaps, Scatter plots with histograms.		
	5	A scatter plot matrix, Area plots.		
References	1 Thomas Nield, Essential Math for Data Science - Take Control of Your Data with Fundamental Linear Algebra, Probability, and Statistics, O'Reilly Media, 2022 2 Wes McKinney, Python for Data Analysis_ Data Wrangling with pandas, NumPy, and Jupyter-O'Reilly Media, Third Edition, 2022 3 Fabio Nelli, Python Data Analytics- With Pandas, NumPy, and Matplotlib, Apress, Second Edition, 2018 4 https://www.kaggle.com/datasets/yasserh/titanic-dataset 5 https://www.w3schools.com/datascience/ds_python.asp 6 https://realpython.com/python-for-data-analysis/ 7 https://www.geeksforgeeks.org/data-science-with-python-tutorial/ 8 https://learn.microsoft.com/en-us/training/modules/explore-analyze-data-with-python/1-introduction 9 https://onlinecourses.nptel.ac.in/noc24_cs54/preview 10 https://onlinecourses.nptel.ac.in/noc20_cs46/preview			

Note: For detailed understanding of the topics given in Module II, additional reference 1 can also be used, though it is not very essential.

Roadmap:

Being a practice-oriented course, the teachers may introduce the students to more problems so as to familiarize them with the tools in which they have been trained through this course. Many good examples on how to use these in real life situations can be found in Chapter 13 of additional reference 2 and the URLs provided in the additional references section.

Mapping of COs with PSOs and POs :

	PSO 1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	3	3	1	3	2	3	3	1	1	1
CO 2	3	2	3	2	3	2	1	1	1	1	1
CO 3	3	2	2	1	3	1	3	3	1	-	1

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Quiz	End Semester Examinations
CO 1	√	√		√
CO 2	√	√	√	√
CO 3	√		√	√

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Internal Exam
- Assignment
- Quiz
- End Semester Examinations